Name: ________________________ (   ) CT Group: 23S0 ____

RAFFLES INSTITUTION

2023 YEAR 6 JULY COMMON TEST

H2 PHYSICS 9749

Section B

INSTRUCTIONS TO CANDIDATES

Write your name, index number and CT Group.

Write your answers to Section B in the spaces provided in this booklet.

For Examiner's Use		
Section A	MCQ	/ 15
Section B	16	/ 12
	17	/ 12
	18	/ 11
	19	/ 10
	20	/ 8
	21	/ 11
	22	/ 16
Deductions		
Total		/ 95

Section B

16 Three points A, B and C are at the corners of an equilateral triangle with sides of length 1.5 m as shown in Fig. 16.1.

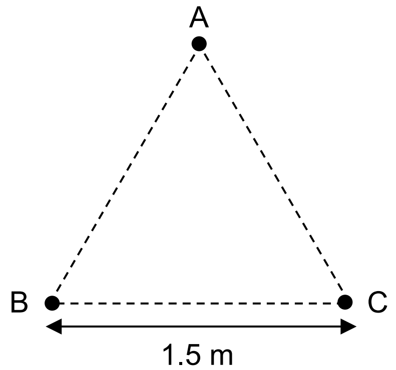

Fig. 16.1

(a) Three charges $q_1 = 4.5$ pC, $q_2 = 4.5$ pC and $q_3 = -2.4$ pC are at rest at infinity. They are brought to points A, B and C respectively in sequence.

Determine the work done by an external force to bring:

(i) q_1 to point A

work done = ... J [1]

(ii) q_2 to point B with q_1 fixed at point A

work done = ... J [1]

(iii) q_3 to point C with q_1 fixed at point A and with q_2 fixed at point B.

work done = ... J [2]

(b) Hence, calculate the electric potential energy of the system of three charges when they are at points A, B and C.

electric potential energy = ... J [1]

(c) q_3 is now released from rest while q_1 and q_2 remain fixed at points A and B respectively.

(i) On Fig. 16.1, draw an arrow to indicate the direction of the electric force acting on charge q_3 at point C.

[1]

(ii) The mass of q_3 is 1.7×10^{-20} kg.

Calculate the speed of q_3 as it passes the line joining points A and B.

speed = ... m s^{-1} [3]

(iii) Describe the motion of q_3 after release.

....

....

....

....

....

.... [3]

17 **(a)** A battery of e.m.f. 12.0 V with negligible internal resistance is connected to two uniform resistance wires X and Y as shown in Fig. 17.1.

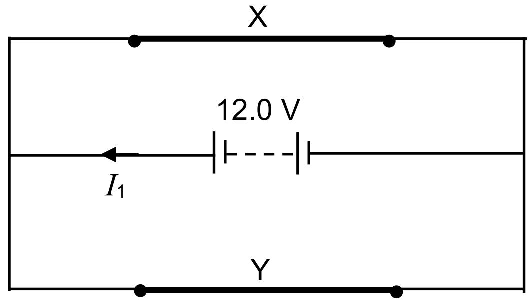

Fig. 17.1

X and Y are of the same length and diameter. The resistivity of Y is twice that of X.

(i) X and Y are known to be *ohmic* resistors.

Explain what is meant by *ohmic*.

....

.... [1]

(ii) The variation with potential difference V_1 of current I for wire X is shown in Fig. 17.2.

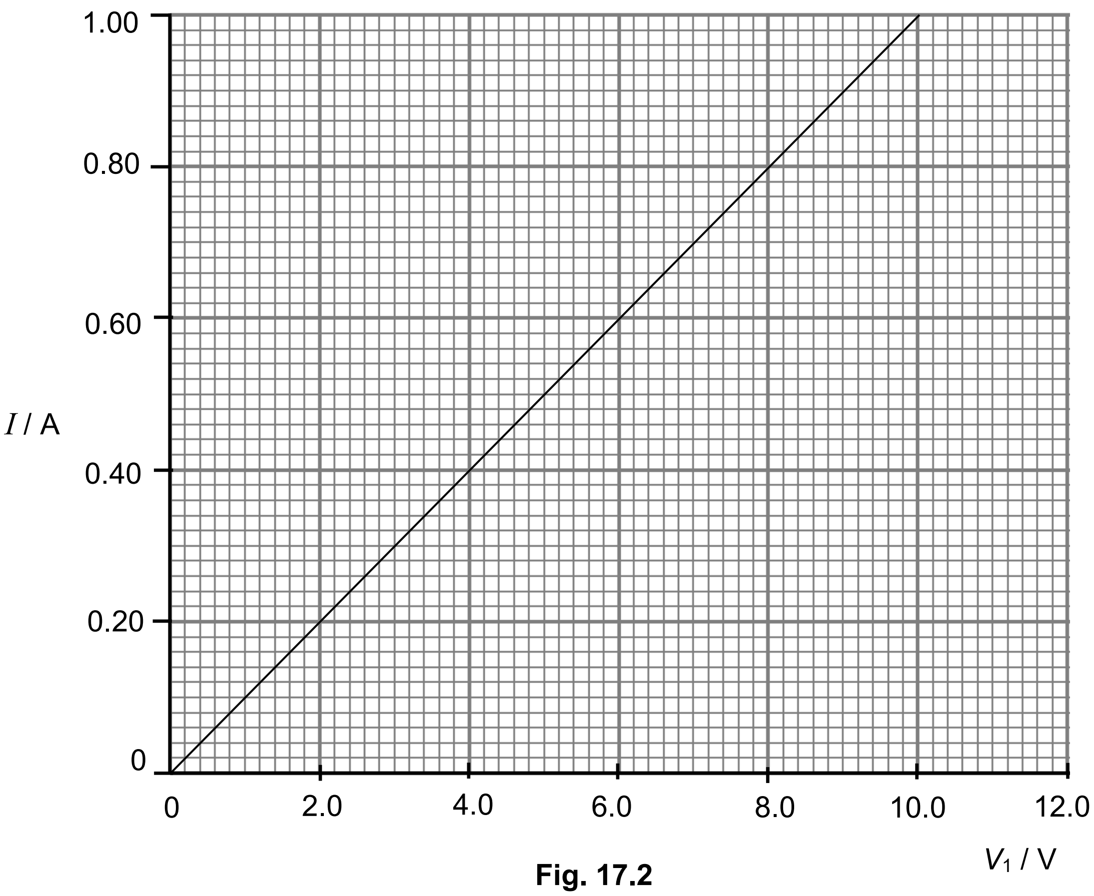

Fig. 17.2

On Fig. 17.2, draw the variation with potential difference V_1 of current I for wire Y.
[1]

(iii) Determine the current I_1 supplied by the battery.

I_1 = ... A [2]

(iv) X and Y are now connected to the 12.0 V battery as shown in Fig. 17.3. One lead of an ideal voltmeter is connected to point D and the other lead is connected to a movable contact.

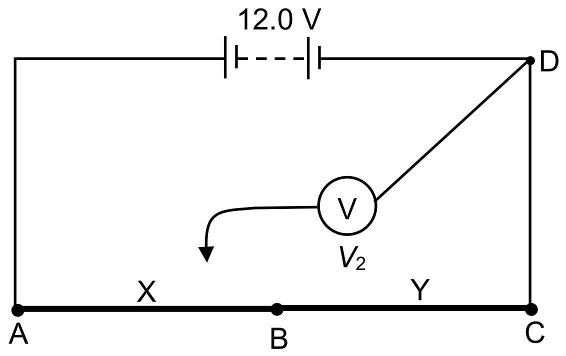

Fig. 17.3

On Fig. 17.4, draw the variation of the reading V_2 on the voltmeter as the movable contact is moved from A to C.

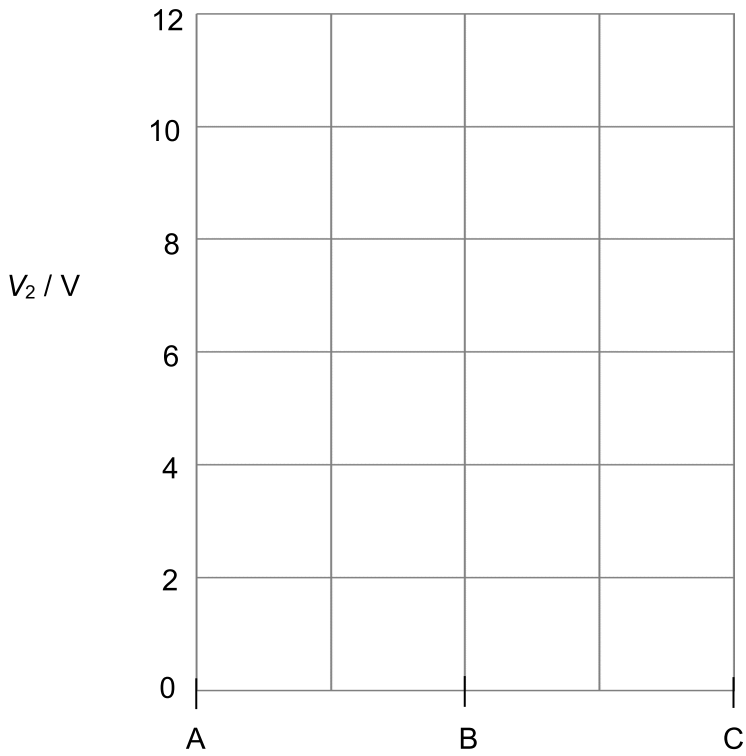

Fig. 17.4

[2]

(b) X and Y are now connected to a 6.0 V battery that is assumed to have negligible internal resistance. The circuit is used to determine the e.m.f. of a cell E as shown in Fig. 17.5.

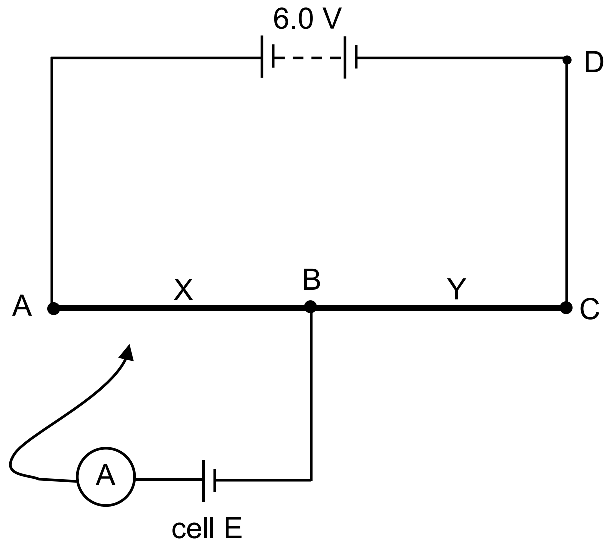

Fig. 17.5

It is observed that when the movable contact is moved along A to B, the ammeter always displays a non-zero reading. Hence, the value of the e.m.f. of E cannot be determined.

(i) Explain why the ammeter always displays a non-zero reading when the movable contact is moved along AB.

None of the apparatus is faulty.

....

.... [1]

(ii) Cell E is now connected to the circuit as shown in Fig. 17.6. A zero reading on the ammeter is obtained when the movable contact is at point J. The length of JC is 0.625 times the length L_Y of wire Y.

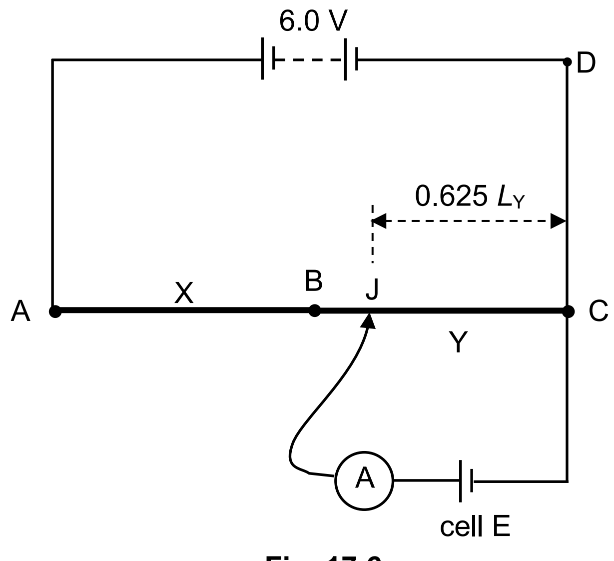

Fig. 17.6

Determine the value of the e.m.f. of cell E.

e.m.f. = ... V [1]

(c) In reality, both the 6.0 V battery and cell E have non-negligible internal resistance.

(i) State and explain the change, if any, this would make to your answer in **(b)(ii)**.

....

....

....

....

....

.... [3]

(ii) The setup in Fig. 17.5 can be modified to determine the internal resistance of cell E.

On Fig. 17.7, complete the circuit diagram to show the setup, using **only** the apparatus in Fig. 17.5, a switch and additional connecting wires if necessary. Label your diagram clearly.

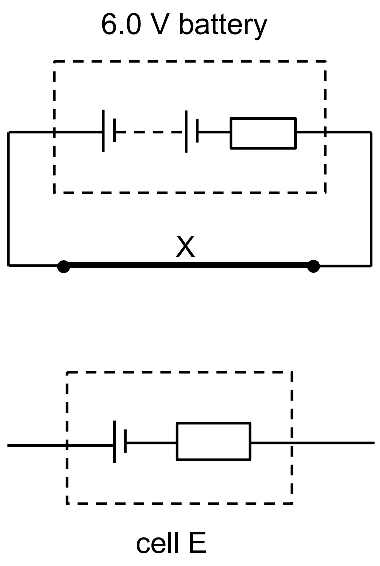

Fig. 17.7

[1]

18 **(a)** Define *magnetic flux density*.

....

....

....

.... [2]

(b) Show that the magnetic flux density at the centre of a flat circular coil of 400 turns and diameter 25.0 cm carrying a current of 3.65 A is 7.34×10^{-3} T.

[1]

(c) Two flat coils, identical to that in **(b)** and carrying a current of 3.65 A, are fixed so that their planes are parallel and are separated by a short distance. Fig. 18.1 provides two different views of the setup.

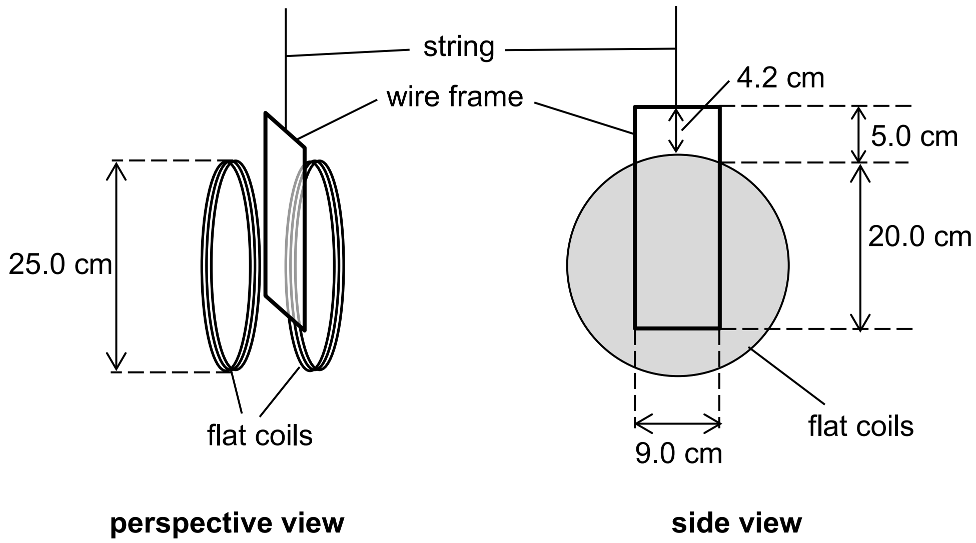

Fig. 18.1

The currents in the two coils produce magnetic field in the same direction in the region between the coils.

It may be assumed that the magnetic field in the region between the two coils is uniform with magnetic flux density equal to twice that calculated in **(b)**, while the magnetic field outside the region is negligible.

A rectangular wire frame suspended from a string is placed in the region of the magnetic field such that the upper 5.0 cm of the frame remains outside the magnetic field. The plane of the wire frame is parallel to the planes of the two coils.

The mass of the wire frame is 0.808 g.

(i) Calculate the tension in the string.

tension = ... N [1]

(ii) There is a current of 6.00 A in the wire frame such that the wire frame experiences a magnetic force acting vertically downward.

Calculate the magnitude of the magnetic force.

magnetic force = ... N [2]

(d) The wire frame is now rotated through an angle θ about the vertical axis passing through its centre.

On Fig. 18.2, sketch the variation with θ of the tension in the string for values of θ from 0° to 360°.

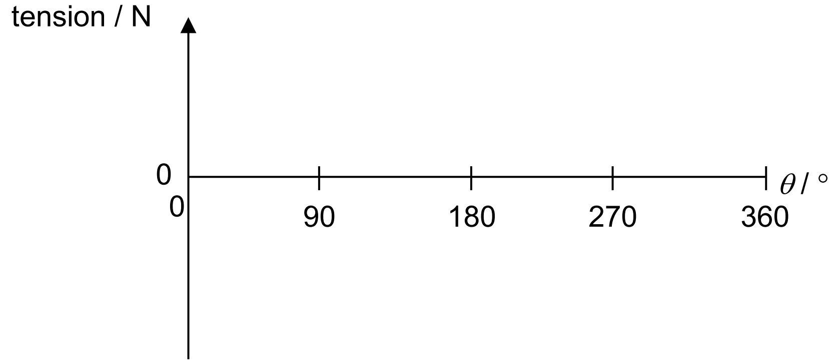

Fig. 18.2

[2]

(e) Calculate the magnitude of the magnetic torque acting on the wire frame when θ = 90°.

magnetic torque = ... N m [3]

19 **(a)** A straight wire PQ of length 0.30 m is pulled across a uniform magnetic field of magnetic flux density 0.024 T at a uniform velocity *v* at an angle of 70° to PQ as shown in Fig. 19.1.

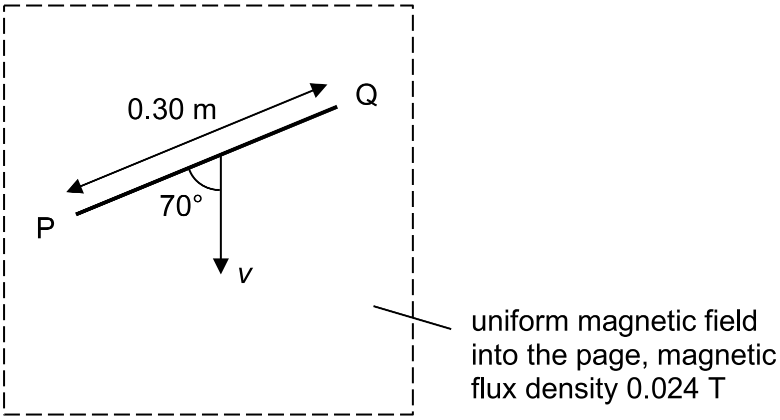

Fig. 19.1

The direction of the magnetic field is into the page.

(i) A potential difference develops across the two ends of the wire.

State and explain which end of the wire has a higher potential.

....

....

....

.... [2]

(ii) The potential difference across PQ is 0.013 V.

Calculate *v*.

v = ... m s^{-1} [3]

(b) The straight wire is now replaced with a wire loop WXYZ that forms the shape of a trapezium. The wire loop is pulled at a uniform velocity *v* in the direction parallel to YZ as shown in Fig. 19.2.

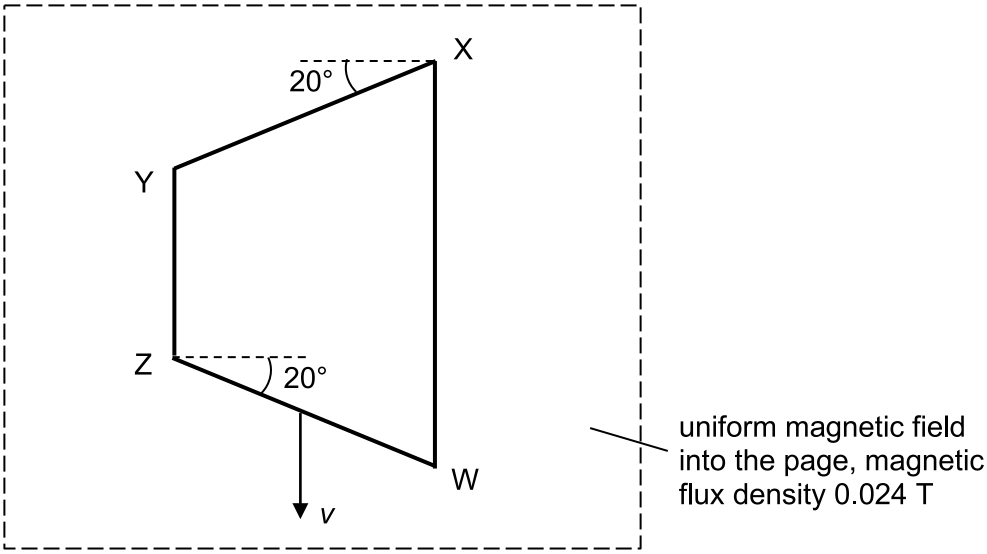

Fig. 19.2

(i) State and explain the magnitude of the induced e.m.f. in the wire loop when the wire loop is within the magnetic field.

....

....

....

.... [2]

(ii) The wire loop moves out of the region of magnetic field at a constant velocity.

At t = 0, the loop is completely inside the field.
At t = t_1, the loop starts to exit the field.
At t = t_2, the loop is just completely out of the field.

On Fig. 19.3, sketch a graph to show the variation with time t of induced e.m.f. E in the wire loop from t = 0 to t = t_2.

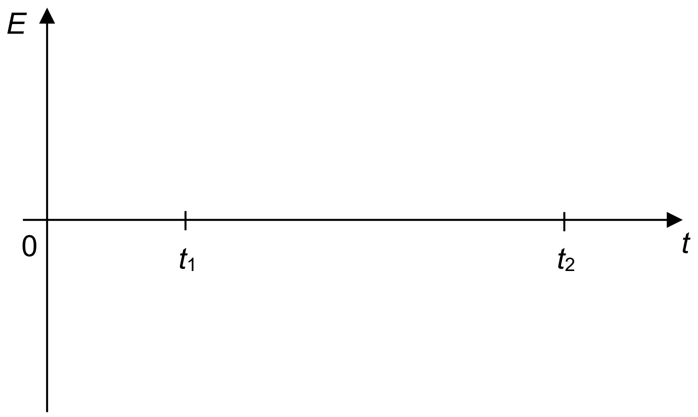

Fig. 19.3

[2]

(iii) Explain why an external force is needed for the wire loop to move at a constant velocity from t_1 to t_2.

....

.... [1]

20 An a.c. power supply is connected to a 28 Ω resistor as shown in Fig. 20.1.

Fig. 20.1

Fig. 20.2 shows the variation with time t of the voltage V_1 of the power supply for one cycle of the voltage.

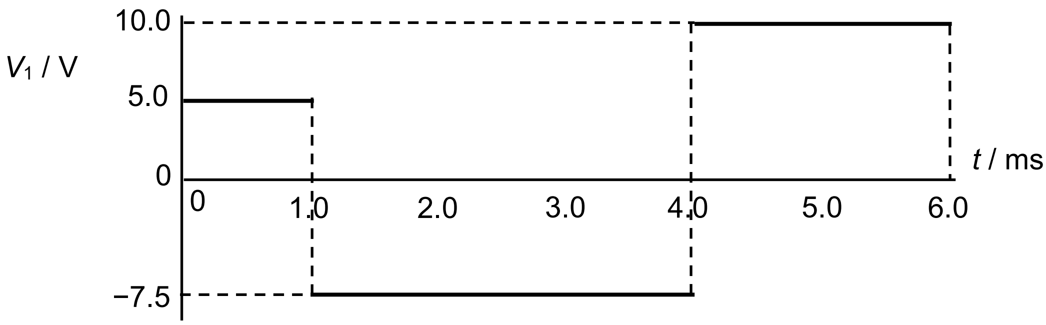

Fig. 20.2

(a) Determine

(i) the mean voltage $\langle V_1 \rangle$ across the resistor

$\langle V_1 \rangle$ = ... V [2]

(ii) the root-mean-square voltage V_{rms}

V_{rms} = ... V [2]

(iii) the mean power $\langle P_1 \rangle$ dissipated in the resistor.

$\langle P_1 \rangle$ = ... W [2]

(b) The a.c. power supply is replaced by another power supply. The voltage V_2 of the new power supply is represented by the equation

$$V_2 = 12\sin(100\pi t),$$

where V_2 is measured in volts and time t is in seconds.

A diode is now connected in series with the 28 Ω resistor as shown in Fig. 20.3.

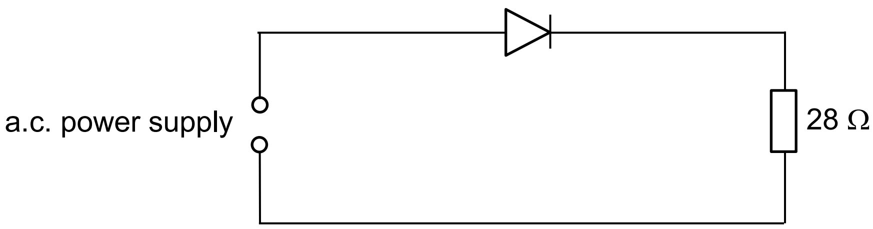

Fig. 20.3

Determine the mean power $\langle P_2 \rangle$ dissipated in the 28 Ω resistor.

$\langle P_2 \rangle$ = ... W [2]

21 **(a)** State what is meant by a *photon*.

....

....

....

.... [2]

(b) A narrow beam of laser light is incident perpendicularly on the surface of a plane mirror and all the photons are reflected along the same path.

The beam has a circular cross-section of radius 0.70 mm.
The laser light has wavelength 650 nm and intensity 4.5×10^{3} W m^{-2}.

(i) Determine

1. the momentum of a photon of the laser light

momentum = ... kg m s^{-1} [2]

2. the energy of a photon of the laser light.

energy = ... J [2]

(ii) Show that the rate of incidence of photons on the mirror is 2.3×10^{16} s^{-1}.

[2]

(iii) Calculate the magnitude of the force that is exerted by the laser light on the surface of the mirror.

magnitude of force = ... N [3]

22 ITER ("The Way" in Latin) is the world's largest nuclear fusion experiment. In southern France, 35 nations are collaborating to build the world's largest tokamak, a magnetic fusion device that has been designed to prove the feasibility of nuclear fusion as a large-scale and carbon-free source of energy based on the same principle that powers the Sun.

Some data for a few nuclear fusion reactions are given in Fig. 22.1.

reaction	equation	energy released / MeV
deuterium-deuterium	${}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + {}^{1}_{0}n$	3.27
deuterium-deuterium	${}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{1}H + {}^{1}_{1}H$	4.03
deuterium-helium-3	${}^{2}_{1}H + {}^{3}_{2}He \rightarrow {}^{4}_{2}He + {}^{1}_{1}H$	18.3
deuterium-tritium	${}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$	

Fig. 22.1

Fig. 22.2 shows the variation with temperature of the fusion reaction rate.

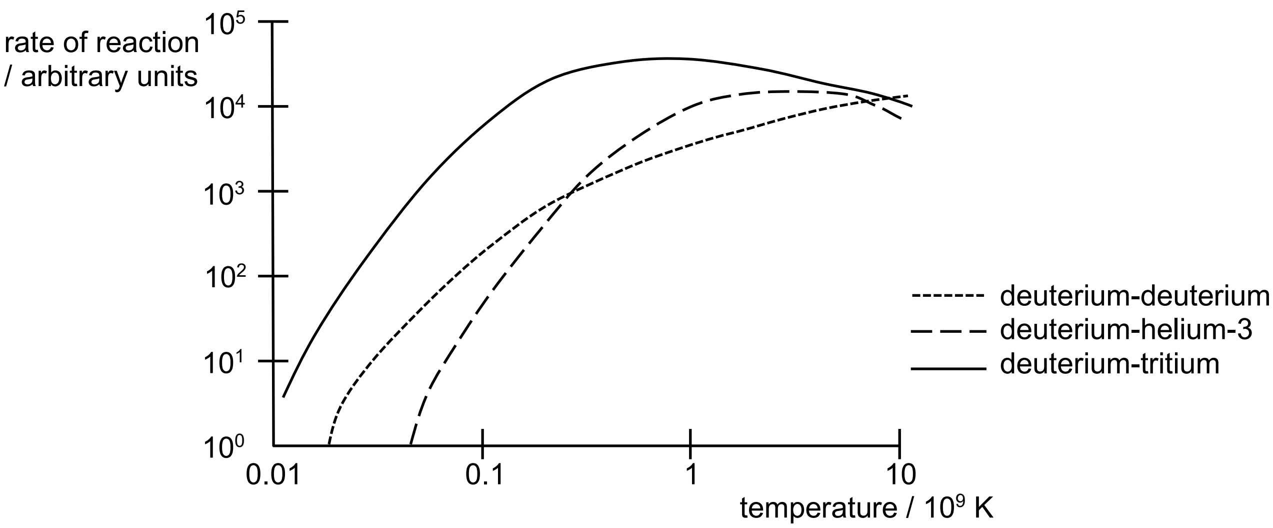

Fig. 22.2

ITER's stated purpose is scientific research. As a research reactor, the thermal energy generated will not be converted to electricity, but simply vented.

ITER will use a mix of deuterium-tritium for its fusion. The first isotope, deuterium, can be extracted from seawater, which means it is a nearly inexhaustible resource. The second isotope, tritium, only occurs in trace amounts in nature and the estimated world's supply is just 20 kilograms per year, which is insufficient for power plants.

Owing to this limited supply of tritium on Earth, a key component of the ITER reactor design is the breeding blanket. This component serves to produce tritium through nuclear reactions with neutrons.

A concrete cylinder with walls 3.5 m thick called the bioshield surrounds the ITER reactor to prevent X-rays, gamma rays, and free neutrons from escaping the reactor.

The free neutrons produced through fusion, known as neutron radiation, can be hazardous. Free neutrons can react with stable nuclides to produce radioactive nuclides. For example, they may react with the phosphorus atoms in the human body. Phosphorus is present in every cell of the human body, with most of it found in the bones and teeth.

Slow moving neutrons react with nuclei of phosphorus-31 to form the phosphorus-32 isotope. The reaction is represented by the equation

$$^{31}_{15}\mathrm{P} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{32}_{15}\mathrm{P} + \gamma$$

Phosphorus-31 is stable but phosphorus-32 is radioactive and decays by beta decay with a half-life of 14.3 days.

Data for some masses are given in Fig. 22.3. Other than the neutron, the mass provided is the atomic mass.

		mass / u
hydrogen-1	$^{1}_{1}\mathrm{H}$	1.00783
neutron	$^{1}_{0}\mathrm{n}$	1.00866
deuterium	$^{2}_{1}\mathrm{H}$	2.01355
tritium	$^{3}_{1}\mathrm{H}$	3.01550
helium-3	$^{3}_{2}\mathrm{He}$	3.01603
helium-4	$^{4}_{2}\mathrm{He}$	4.00150
phosphorus-31	$^{31}_{15}\mathrm{P}$	30.97376
phosphorus-32	$^{32}_{15}\mathrm{P}$	31.97391

Fig. 22.3

(a) State what is meant by nuclear fusion.

....

.... [1]

(b) Use data from Fig. 22.3 to determine, in MeV, the energy released in the deuterium-tritium reaction.

energy released = ... MeV [3]

(c) Use Fig. 22.2 to explain why ITER uses the deuterium-tritium reaction.

....

.... [1]

(d) Explain why thick concrete is required to stop neutrons from escaping the reactor.

....

.... [1]

(e) **(i)** State what is meant by binding energy of a nucleus and how it is related to the mass defect.

....

....

....

.... [2]

(ii) Use data from Fig. 22.3 to determine, to three decimal places, the binding energy per nucleon, in MeV, of phosphorus-31.

binding energy per nucleon = ... MeV [3]

(f) Phosphorus-32 ($^{32}_{15}P$) emits a beta particle to become sulfur (S).

(i) Complete the nuclear decay equation, including all the decay products.

$^{32}_{15}P \rightarrow$ [3]

(ii) Explain why there is a large variation in the kinetic energy of the beta particle.

....

.... [1]

(iii) Determine the probability of decay of a phosphorus-32 nucleus in a time of 2.0 hour.

probability of decay = ... [1]